

Basics 48: Energetic-Particle Distributions

VI - Kinetic theory and energetic particles

Fusion-alpha birth, collisional slowing down, neutral-beam anisotropy, radial gradients, normalization, and numerical sampling

Textbook draft, version 1.0. Alfvén-eigenmode stability is controlled not only by the total fast-ion density but by its distribution in radius, energy, and pitch. This chapter constructs reduced energetic-particle distributions for fusion-born alpha particles and neutral-beam ions, derives the classical slowing-down form and its normalization, explains anisotropy and pressure moments, and defines data contracts suitable for Modules 2, 3, and 5.

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1 Learning objectives

After completing this note, the reader should be able to:

- Distinguish birth, slowing-down, and loss-modified energetic-particle distributions.
- Normalize isotropic distributions in speed or energy coordinates.
- Derive the classical $1/(v^3 + v_c^3)$ slowing-down shape.
- Represent pitch anisotropy and radial gradients.
- Compute density and pressure moments of a reduced distribution.
- Design importance-sampling distributions for orbit and resonance calculations.

2 Prerequisites and reading strategy

Recommended prerequisites are Basics 1, 19, 39, 41, 42, and 44. Every formula must be paired with its integration measure; otherwise normalization is ambiguous.

3 Coordinates for energetic particles

Useful variables include speed v , kinetic energy

$$E = \frac{mv^2}{2}, \quad (3.1)$$

pitch cosine

$$\xi = \frac{v_{\parallel}}{v} \in [-1, 1], \quad (3.2)$$

and a magnetic-moment pitch coordinate

$$\lambda = \frac{\mu B_{\text{ref}}}{E}. \quad (3.3)$$

For a gyrotropic distribution,

$$d^3v = 2\pi v^2 dv d\xi. \quad (3.4)$$

Since $dv = dE/(mv)$,

$$d^3v = \frac{2\pi v}{m} dE d\xi. \quad (3.5)$$

A distribution density per unit E and ξ therefore differs from a Cartesian velocity-space f by this Jacobian.

4 Fusion-alpha birth source

The D–T reaction produces a 3.5 MeV alpha particle and a 14.1 MeV neutron. A reduced local alpha source is

$$S_{\alpha}(\psi, E, \xi) = n_D(\psi)n_T(\psi)\langle\sigma v\rangle_{DT}\delta(E - E_{\alpha})g_{\xi}(\xi), \quad (4.1)$$

For the distinct D and T reactants no factor of 1/2 is used; that combinatorial factor is required only for reactions between identical species. For isotropic birth,

$$g_{\xi}(\xi) = \frac{1}{2}, \quad -1 \leq \xi \leq 1. \quad (4.2)$$

The spatial source is strongly peaked where density and temperature make the fusion reactivity largest.

5 Classical slowing-down balance

At high speed, alpha particles lose energy primarily to electrons; at lower speed, ion drag becomes important. A simplified isotropic steady kinetic balance in speed is

$$\frac{1}{v^2} \frac{d}{dv} \left[v^2 \dot{v} f(v) \right] + S_0 \frac{\delta(v - v_{\alpha})}{4\pi v^2} = 0. \quad (5.1)$$

A standard drag model has

$$\dot{v} = -\frac{\nu_s}{v^2}(v^3 + v_c^3), \quad (5.2)$$

where v_c is the critical speed separating electron-dominated and ion-dominated slowing. Below the birth speed and away from the delta source, the speed-space flux is constant, giving

$$\boxed{f_s(v) = \frac{C}{v^3 + v_c^3} H(v_\alpha - v)}. \quad (5.3)$$

This is a non-Maxwellian distribution with a broad energetic tail.

6 Normalization of the slowing-down distribution

For an isotropic population of density n_f ,

$$n_f = 4\pi \int_0^{v_\alpha} f_s(v) v^2 dv. \quad (6.1)$$

Using

$$\int \frac{v^2}{v^3 + v_c^3} dv = \frac{1}{3} \ln(v^3 + v_c^3), \quad (6.2)$$

we obtain

$$n_f = \frac{4\pi C}{3} \ln \left[1 + \left(\frac{v_\alpha}{v_c} \right)^3 \right]. \quad (6.3)$$

Therefore

$$\boxed{C = \frac{3n_f}{4\pi \ln[1 + (v_\alpha/v_c)^3]}}. \quad (6.4)$$

This normalization is valid for the six-dimensional velocity distribution $f_s(v)$ integrated with $4\pi v^2 dv$.

7 Energy-space form

Define an energy density $F_E(E)$ such that

$$n_f = \int_0^{E_\alpha} F_E(E) dE. \quad (7.1)$$

For an isotropic velocity distribution,

$$F_E(E) = 4\pi v^2 f(v) \frac{dv}{dE} = \frac{4\pi v}{m} f(v). \quad (7.2)$$

Confusing $f(v)$ with $F_E(E)$ changes both units and shape. Machine-readable outputs should store coordinate names and measures explicitly.

8 Neutral-beam distributions

Neutral-beam injection produces a directional source with finite energy spread. A simple model is

$$S_{\text{NBI}}(\psi, E, \xi) = S_0 g_r(\psi) \exp\left[-\frac{(E - E_b)^2}{2\Delta E^2}\right] \exp\left[-\frac{(\xi - \xi_b)^2}{2\Delta \xi^2}\right]. \quad (8.1)$$

Co-current tangential injection has ξ_b near one, while perpendicular injection has smaller $|\xi_b|$. Ionization geometry and orbit topology make the true source nonseparable; the Gaussian model is only a controlled benchmark.

9 Radial profiles and gradients

A reduced separable distribution may be written

$$f_f(\psi, E, \xi) = n_f(\psi) \hat{f}_E(E; \psi) \hat{f}_\xi(\xi; \psi), \quad (9.1)$$

where each hatted factor is normalized under its stated measure. Example radial profiles include

$$n_f(s) = n_{f0}(1 - s^a)^b, \quad 0 \leq s \leq 1. \quad (9.2)$$

The logarithmic gradient

$$\frac{d \ln n_f}{ds} \quad (9.3)$$

is often more relevant to wave drive than the absolute density. Alfvén-eigenmode transport tends to flatten resonant gradients.

10 Pressure and anisotropy moments

For a gyrotropic distribution,

$$p_{\parallel} = m \int v_{\parallel}^2 f d^3v, \quad (10.1)$$

$$p_{\perp} = \frac{m}{2} \int v_{\perp}^2 f d^3v. \quad (10.2)$$

An isotropic population has $p_{\parallel} = p_{\perp} = p_f$. A beam-like distribution can have strong anisotropy, changing both equilibrium pressure and kinetic coupling to modes. The fast-ion beta is

$$\beta_f = \frac{2\mu_0 p_f}{B^2} \quad (10.3)$$

for isotropic scalar pressure.

11 Loss cones and orbit filtering

The source distribution is not necessarily the confined distribution. Prompt orbit losses can remove selected pitch and radial regions. A reduced loss-cone factor might be

$$L(\psi, E, \xi) \in [0, 1], \quad f_{\text{conf}} = L f_{\text{source}}. \quad (11.1)$$

Module 3 should estimate L from orbit following rather than assuming it. This creates the coupling

$$S_f \longrightarrow \text{orbits} \longrightarrow f_{\text{conf}} \longrightarrow \text{mode drive}. \quad (11.2)$$

12 Sampling and marker weights

If markers are sampled from $g(\psi, E, \xi)$, an integral is estimated by weights $w = f/g$. Uniform sampling is inefficient when resonance occupies a narrow phase-space region. Useful strategies include:

- sample proportional to particle density for global moments;
- oversample high-energy tails for wall-loss power;
- oversample near resonance surfaces for kinetic drive;
- use stratified sampling in pitch and radius to capture rare trapped or loss orbits.

Weights and marker-density normalization must be stored with the particle data.

13 Uncertainty and model hierarchy

A parameterized slowing-down distribution is appropriate for early verification, but predictive studies require collisions, sources, orbit loss, radial transport, and possibly time dependence:

$$\frac{\partial f_f}{\partial t} + \dot{\mathbf{Z}} \cdot \nabla_{\mathbf{Z}} f_f = C[f_f] + S_f - L_f. \quad (13.1)$$

The hierarchy should separate uncertainty in source rate, critical speed, radial profile, pitch dependence, and loss model.

14 Connection to the Kinetic MHD-PINN project

Module 2 owns the source and distribution representation. Module 3 applies orbit filtering and computes frequencies. Module 5 evaluates resonant drive using gradients of f_f . Module 6 evolves

redistribution. The shared-state contract should include coordinate definitions, Jacobian, units, normalization, species, reference magnetic field, and time stamp.

15 Computational laboratory

1. Normalize the classical slowing-down distribution by quadrature and compare with the analytic constant C .
2. Transform it from speed to energy coordinates and verify that both integrals yield n_f .
3. Sample markers by inverse or rejection sampling and recover density and pressure moments.
4. Add a pitch Gaussian and measure $p_{\parallel}/p_{\perp}$.
5. Apply a synthetic loss cone and quantify changes in density, energy, and resonant population.

16 Summary

Energetic-particle stability depends on the full distribution in space, energy, and pitch. Fusion alphas are born nearly monoenergetically and relax toward a classical slowing-down form; neutral beams produce anisotropic sources. Correct Jacobians and normalization are indispensable. Radial and phase-space gradients supply free energy for Alfvén eigenmodes, while orbit loss and transport reshape those gradients.

17 Exercises

1. Derive the normalization constant for $f_s(v) = C/(v^3 + v_c^3)$.
2. Derive the energy-coordinate density $F_E(E)$.
3. Show that an isotropic distribution satisfies $p_{\parallel} = p_{\perp}$.
4. Explain why total fast-ion density alone cannot determine AE stability.
5. Design a JSON schema that prevents ambiguity between $f(v)$ and $F_E(E)$.

Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
f_f	Energetic-particle distribution	Fast-ion phase-space density under a stated measure.
E_α	Alpha birth energy	Approximately 3.5 MeV for D–T fusion.
v_c	Critical speed	Speed separating electron- and ion-dominated collisional slowing.
ξ	Pitch cosine	$v_{\parallel}/v$.
λ	Magnetic pitch coordinate	Normalized magnetic moment coordinate.
β_f	Fast-ion beta	Ratio of fast-ion pressure to magnetic pressure.

18 References

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2. R. B. White, *The Theory of Toroidally Confined Plasmas*.
3. J. Wesson, *Tokamaks*.
4. L. Chen and F. Zonca, energetic-particle and Alfvén-wave review literature.